Chapter 1: The Real and Complex Number Systems

Section: Introduction

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

Example 1.1: Irrationality of $\sqrt{2}$

Remark 1.2: Gaps in the Rationals

Definitions 1.3–1.4: Set-Theoretic Terminology

Summary

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- Analysis requires a precisely defined number concept
- The rational numbers $\mathbb{Q}$ are *not enough*
- Goal: understand the gaps in $\mathbb{Q}$ that force us to build $\mathbb{R}$

Main ideas in this section:

- $\sqrt{2}$ is irrational (Example 1.1)
- The rationals have gaps that the reals fill (Remark 1.2)
- Basic set-theoretic language (Definitions 1.3–1.4)

The Problem with Rationals

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The sequence $1,\ 1.4,\ 1.41,\ 1.414,\ 1.4142,\ \ldots$ "tends to $\sqrt{2}$"

But what *is* $\sqrt{2}$ if we only have $\mathbb{Q}$?

Example 1.1: Irrationality of $\sqrt{2}$

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Technique: Proof by contradiction

Assume $p = m/n$ with $m, n \in \mathbb{Z}$, not both even.

Then (1) gives:

$$m^2 = 2n^2 \tag{2}$$

Example 1.1: $\sqrt{2}$ Is Irrational — The Contradiction

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Both m and n are even — contradicts our assumption!

$\boxed{\text{No rational } p \text{ satisfies } p^2 = 2.}$

Example 1.1: The Sets A and B

Define two subsets of positive rationals:

$$A = \{p \in \mathbb{Q}^+ : p^2 < 2\}, \qquad B = \{p \in \mathbb{Q}^+ : p^2 > 2\}$$

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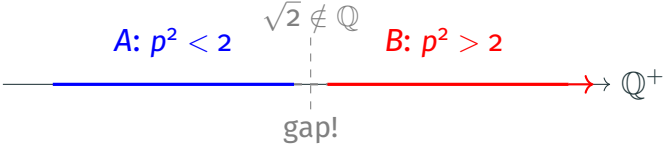

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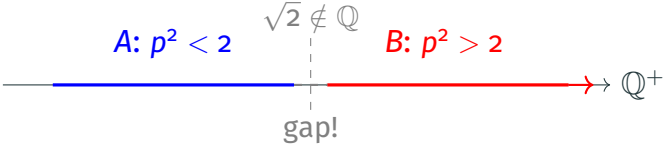

Claim: A has no largest element, and B has no smallest element.

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$$q = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} \tag{3}$$

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Then:

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2} \tag{4}$$

Key observation: $q^2 - 2$ has the same sign as $p^2 - 2$.

Example 1.1: Completing the Argument

Case 1: $p \in A$ (so $p^2 < 2$, i.e., $p^2 - 2 < 0$)

- From (3): $q > p$ (subtracting a negative number)
- From (4): $q^2 - 2 < 0$, so $q^2 < 2$
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Case 2: $p \in B$ (so $p^2 > 2$, i.e., $p^2 - 2 > 0$)

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$\boxed{A \text{ has no largest element, } B \text{ has no smallest element.}}$

Remark 1.2: Gaps in the Rationals

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The rational number system has gaps, despite the fact that between any two rationals there is another:

$$r < s \implies r < \frac{r+s}{2} < s$$

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The real number system fills these gaps — the principal reason $\mathbb{R}$ is fundamental to analysis.

Definitions 1.3–1.4: Set-Theoretic Terminology

Definition 1.3: Basic Set Notation

Definition 1.3: Set Membership

- $x \in A$: x is a *member* (element) of A
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Definition 1.3 (cont.): Subsets

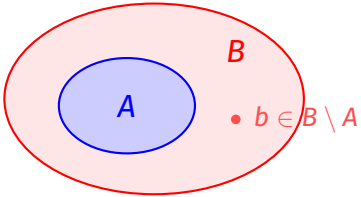
- $A \subset B$: every element of A is an element of B
  (A is a *subset* of B)
- If also $\exists\, b \in B$ with $b \notin A$: A is a proper subset of B
- $A \subset A$ for every set A

Definition 1.3: Set Equality

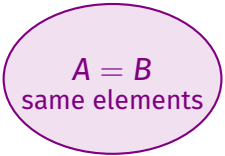
Definition 1.3 (cont.): Equality of Sets

$$A = B \iff A \subset B \text{ and } B \subset A$$

Otherwise, $A \neq B$.



$A \subset B$ (proper)



$A \subset B$ and $B \subset A$

Definition 1.4: The Rationals $\mathbb{Q}$

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Throughout Chapter 1, the set of all rational numbers is denoted by Q.

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What comes next:

- *Ordered sets* — formalize the notion of "$<$" (Section 2)
- *Fields* — formalize arithmetic operations (Section 3)
- Build $\mathbb{R}$ from $\mathbb{Q}$ to fill the gaps (Section 4)

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- **Remark 1.2:** Despite density, $\mathbb{Q}$ has gaps; $\mathbb{R}$ fills them

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- Analysis needs a precise number system; $\mathbb{Q}$ is *not* sufficient
- **Example 1.1:** $\sqrt{2}$ is irrational (proof by contradiction)
- The sets $A = \{p \in \mathbb{Q}^+ : p^2 < 2\}$ and $B = \{p \in \mathbb{Q}^+ : p^2 > 2\}$ have a gap — A has no max, B has no min
- **Remark 1.2:** Despite density, $\mathbb{Q}$ has gaps; $\mathbb{R}$ fills them
- **Definitions 1.3–1.4:** Set membership ($\in$), subsets ($\subset$), equality, and the notation Q for the rationals

Coming up next: Ordered Sets — formalizing the relation "$<$"